

BGE-Attention: A Probabilistic Prediction Framework for Social Media Comment Popularity Based on Cross-Attention Mechanism

Anonymous ACL submission

Abstract

Social media information ecosystems exhibit typical characteristics of complex adaptive systems, where the generation mechanism of comment popularity embeds ontological uncertainty, revealing fundamental limitations of traditional point estimation paradigms. This paper re-examines the popularity prediction problem from an epistemological perspective, arguing that its essence is the estimation of conditional probability distributions with inherent randomness. Based on this insight, we propose the BGE-Attention probabilistic prediction framework: encoding multi-source texts through the BGE pre-trained model, achieving conditional projection in semantic space via Cross-Attention mechanism; designing a log-scale negative log-likelihood loss function to reconstruct scale invariance in error measurement; and proposing MinHash-based temporal-semantic joint features to construct computable representations of the “first-mover vs. follower” effect. Experiments on 270,000 Weibo comments demonstrate that compared to the NG-Boost baseline, our method reduces the Mean Prediction Interval Width (MPIW) by six orders of magnitude (from tens of millions to 3.0050), while improving PICP@95% from 94.30% to 97.75%, achieving Pareto optimality between prediction accuracy and uncertainty calibration in social media scenarios for the first time. Code is available at: https://github.com/LJK666666666/LLM_SU7.

1 Introduction

1.1 Research Background: Popularity Prediction from a Complex Systems Perspective

Social media has evolved into a typical **complex adaptive system**: micro-level interaction behaviors of individual users and macro-level propagation patterns of emergent groups are mutually coupled, forming cross-scale nonlinear dynamics (Liu,

2019). Comment popularity—as a projection of the “content-user-platform” ternary interaction—embeds **ontological uncertainty** in its generation mechanism: identical semantic content at different temporal positions can produce order-of-magnitude differences in propagation effects. This phenomenon has significant application value for public opinion monitoring (Zhang and Zhang, 2020), brand management, and content recommendation (Wu et al., 2019), but also reveals the fundamental limitations of traditional deterministic prediction paradigms.

From an information-theoretic perspective, existing research has fallen into the “**point estimation trap**”: forcibly mapping the inherently stochastic popularity evolution process to a deterministic numerical space. This not only causes semantic deficiency in prediction confidence but fundamentally conflates the boundaries between **epistemic uncertainty** (arising from incomplete model knowledge) and **aleatoric uncertainty** (arising from inherent data randomness).

1.2 Technical Challenges: Systematic Analysis of Triple Bottlenecks

Through in-depth analysis of existing methods, we identify three technical bottlenecks constraining prediction performance:

(1) **Scale distortion effect of loss functions.** Social media data exhibits typical power-law distribution, where a few comments receive massive interactions while most receive sparse engagement. Traditional mean squared error loss shows severe **dimensional bias** in this scenario: excessive sensitivity to absolute errors for high-popularity samples while systematically underestimating relative errors for low-popularity samples. The essence of this deficiency lies in the failure of Euclidean distance metrics to capture the **multiplicative noise characteristics** of social media data—prediction errors should be proportional to true values rather

than absolute.

(2) **Decoupling dilemma between semantic representation and context.** Existing text encoding schemes generally adopt naive “encode-concatenate” strategies, ignoring the **conditional dependency structure** between comment semantics and their context (Weibo posts, parent comments, root comments). Such architectures cannot model the phenomenon of “same comment having vastly different popularity in different contexts,” essentially constituting systematic omission of **pragmatic information**.

(3) **Modeling gap for temporal position effects.** The popularity gap between “first-mover effect” and “follower effect” reveals a long-neglected prior: **the interaction between textual similarity and temporal precedence** constitutes a key explanatory variable for popularity prediction. However, the $O(n^2)$ complexity of brute-force text similarity computation makes this feature impractical at scale.

1.3 Our Contributions: From Tactical Improvements to Paradigm Innovation

Addressing these challenges, this paper proposes the BGE-Attention probabilistic prediction framework, achieving a **paradigm shift** from “point estimation” to “distribution prediction.” Main contributions include:

1. **Prediction paradigm reconstruction:** Proposing the BGE-Attention model with dual prediction heads that simultaneously output mean $\mu(x)$ and standard deviation $\sigma(x)$, achieving joint modeling of predicted values and prediction confidence, explicitly encoding the inherent randomness of prediction targets;
2. **Reconstructing scale invariance in loss functions:** Designing a log-scale negative log-likelihood loss function that achieves equal-weight penalization of cross-magnitude prediction errors by mapping error metrics from original space to log space, while constructing an endogenous uncertainty regularization mechanism;
3. **Conditional projection in semantic space:** Employing Cross-Attention mechanism to establish conditional mapping between “comment semantics-context semantics,” elevating comment representations from “isolated semantics” to “contextualized semantics”;

4. **Constructing temporal-semantic joint feature space:** Proposing MinHash-based duplication features that reduce similarity computation complexity from $O(n^2)$ to $O(nk)$, providing computable mathematical expressions for the “first-mover vs. follower” effect.

2 Related Work

2.1 Social Media Popularity Prediction: From Time Series Modeling to Deep Learning

The research trajectory of social media popularity prediction has undergone paradigm evolution from statistical modeling to deep learning. Early work mainly relied on time series models (Yang and Leskovec, 2011), predicting final popularity by analyzing propagation temporal features (such as early growth rate, peak time), with theoretical foundations in the **Matthew effect** of information propagation—content gaining early attention tends to continue receiving attention. Bandari et al. (2012) studied news article sharing prediction, finding that content features and sources significantly impact performance, revealing the **interaction between content factors and channel factors** in popularity prediction.

In recent years, deep learning methods have achieved significant progress. Deng and Xie (2020) proposed attention-based neural network models to capture complex user-content interactions. However, these methods generally share an **epistemological limitation**: focusing only on point estimation while ignoring uncertainty modeling in predictions. This paper argues that the essence of popularity prediction is the estimation of **conditional probability distributions**, and point estimation paradigms cannot express the model’s confidence in prediction results.

2.2 Pre-trained Language Models: From Bag-of-Words to Context-Aware Representations

Text representation technology has undergone profound transformation from **symbolic space** to **continuous vector space**. Traditional TF-IDF and bag-of-words models map text to sparse high-dimensional vectors, failing to capture word order information and semantic relationships. The emergence of pre-trained language models marks a paradigm revolution in NLP: BERT (Devlin et al., 2019) learns bidirectional contextual representations through masked language modeling, while

RoBERTa (Liu et al., 2019) further optimized pre-training strategies.

BGE (BAAI General Embedding) (Xiao et al., 2023) is a text embedding model designed for Chinese, with core innovation in introducing the **contrastive learning paradigm**—through discriminative training on positive and negative sample pairs, semantically similar texts become closer in embedding space. This paper adopts BGE-base-zh-v1.5 for extracting comment semantic representations and implements **context-aware semantic fusion** through Cross-Attention mechanism, enabling comment representations to dynamically adapt to their conversational context.

2.3 Probabilistic Prediction and Uncertainty Estimation: From Point Estimates to Distribution Outputs

Uncertainty estimation is a research direction with profound theoretical implications in machine learning (Gal, 2016). Bayesian neural networks (Blundell et al., 2015) explicitly model parameter uncertainty by imposing prior distributions on weights; MC Dropout (Gal and Ghahramani, 2016) proposes approximating Bayesian inference through multiple stochastic dropouts, but with significant computational overhead.

NGBoost (Duan et al., 2020) pioneered the **natural gradient boosting** technical route, capable of directly outputting prediction distributions by optimizing parameters of conditional distributions (rather than point predictions). However, this method faces two challenges in social media scenarios: (1) loss functions not designed for long-tail distributions; (2) tree models have difficulty effectively utilizing semantic representations from pre-trained language models. Building upon this, our paper proposes **log-scale loss functions** to reconstruct scale invariance in error metrics, and achieves end-to-end semantic-numerical joint modeling through neural network architectures.

3 Method

3.1 Problem Formalization: From Point Estimation to Distribution Estimation

Given a comment x with its contextual information (associated Weibo post, root comment, parent comment) and statistical features, traditional methods aim to learn the mapping $f : x \mapsto \hat{y}$. This paper redefines the prediction target as estimation of the **conditional probability distribution** $P(y|x)$.

Based on analysis of social media data distribution characteristics, we assume that log-transformed popularity follows a normal distribution:

$$\log(y + c) \sim \mathcal{N}(\mu(x), \sigma^2(x)) \quad (1)$$

where c is a smoothing constant, and $\mu(x)$ and $\sigma(x)$ are the model-predicted **location parameter** (representing prediction center) and **scale parameter** (representing prediction uncertainty), respectively. The core insight of this formalization lies in: **explicitly parameterizing** the uncertainty originally implicit in prediction residuals, enabling the model to output differentiated confidence levels for different samples.

3.2 Model Architecture: Hierarchical Fusion of Multi-Source Semantics

The BGE-Attention model architecture proposed in this paper is shown in Figure 1, adopting a **three-stage “encode-fuse-predict” design** with four functional modules.

3.2.1 Text Encoding Module: Independent Representation of Multi-Source Semantics

BGE-base-zh-v1.5 model is employed for **independent encoding** of four text types, constructing multi-source semantic representation space:

- **Comment Text:** Content of the current comment, carrying core semantic information;
- **Weibo Text:** Main text of the associated Weibo post, providing **topic context**;
- **Root Comment Text:** Content of the comment chain’s root node, reflecting **discussion mainline**;
- **Parent Comment Text:** Content of the directly replied comment, constituting **conversational context**.

Each text yields a 768-dimensional dense vector representation after BGE encoding. **BGE parameters are frozen** to prevent overfitting on downstream tasks while preserving general semantic knowledge acquired during pre-training.

3.2.2 Cross-Attention Fusion Mechanism: Context-Aware Semantic Projection

Distinct from naive vector concatenation strategies, this paper employs Cross-Attention mechanism for **conditioned semantic fusion**. Using the comment

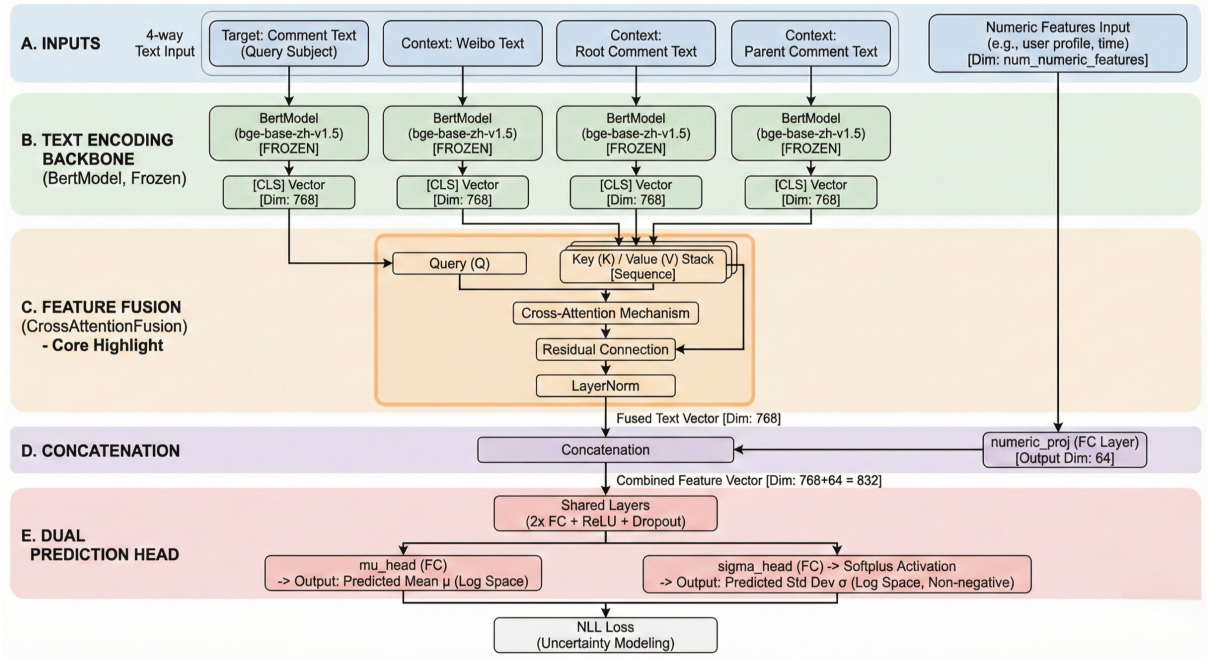


Figure 1: BGE-Attention Model Architecture

vector as Query and context vectors (Weibo, root comment, parent comment) as Key and Value:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

where $Q \in \mathbb{R}^{1 \times 768}$ is the comment vector, $K, V \in \mathbb{R}^{3 \times 768}$ are context vector matrices, and $d_k = 768$ is the scaling factor. The **core advantage** of this mechanism lies in: attention weights are **dynamically computed** based on semantic relevance between the comment and each context, enabling the model to adaptively focus on context information most relevant to the current comment, achieving representation elevation from “isolated semantics” to “contextualized semantics.”

3.2.3 Dual Prediction Heads: Explicit Modeling of Uncertainty

The fused semantic vector is concatenated with numerical features and passed through a **dual prediction head architecture** to output location parameter μ and scale parameter σ separately:

$$h = \text{MLP}([\text{Attention}; \text{NumFeatures}]) \quad (3)$$

$$\mu = W_\mu h + b_\mu \quad (4)$$

$$\sigma = \text{Softplus}(W_\sigma h + b_\sigma) + \epsilon \quad (5)$$

where $\epsilon = 10^{-4}$ is a numerical stability constant, and the Softplus function $\log(1 + e^x)$ ensures

$\sigma > 0$. The design philosophy of dual prediction heads lies in: **decoupling prediction center from prediction uncertainty**, enabling the model to output narrow confidence intervals for “high-certainty simple samples” and wide confidence intervals for “low-certainty difficult samples.”

3.3 Log-Scale NLL Loss Function: Reconstructing Scale Invariance

Social media popularity data approximates normal distribution in log scale, providing statistical justification for adopting Gaussian likelihood. This paper employs log-scale negative log-likelihood (NLL) loss:

$$\mathcal{L} = \frac{1}{2} \log(\sigma^2) + \frac{(\log(y + c) - \log(\mu + c))^2}{2\sigma^2} \quad (6)$$

This loss function embodies threefold design wisdom:

(1) **Scale invariance**: By measuring errors in log space, achieving **equal-weight penalization of relative errors**. As shown in Table 1, $c = 10$ reduces the loss ratio between “predicting 1 when actual is 2” and “predicting 110 when actual is 120” from 4.7 to 1.1, eliminating traditional MSE’s excessive focus on small-value samples.

(2) **Uncertainty regularization**: The $\frac{1}{2} \log(\sigma^2)$ term constructs an endogenous **confidence penalty mechanism**—if the model outputs overly small σ

(overconfidence), it will suffer huge second-term loss when prediction deviations occur; conversely, overly large σ can reduce the second term but significantly increases the first. This forces the model to seek **game equilibrium** between “precise prediction” and “honestly acknowledging uncertainty.”

(3) **Heteroscedastic modeling**: Allowing different samples to have different $\sigma(x)$, enabling the model to output low uncertainty for semantically clear comments (e.g., “thanks for sharing”) and high uncertainty for semantically ambiguous comments (e.g., “view image”).

c	Loss(1,2)	Loss(110,120)	Ratio
1	0.406	0.086	4.7
10	0.087	0.080	1.1

Table 1: Effect of smoothing constant c on loss function scale sensitivity.

3.4 Feature Engineering: Systematic Extraction of Multi-Dimensional Signals

This paper designs four complementary feature categories totaling 17 dimensions, characterizing comment popularity potential from four perspectives: **user profile, text semantics, topic distribution, and temporal position**.

3.4.1 Basic Statistical Features and Text Features

Basic statistical features capture **user influence** and **temporal effects**, while text features characterize **expression intensity** and **domain relevance**, as detailed in Table 2.

Category	Description
Statistical	User total comments, verification, etc.
Text	Length, emoji count, domain keywords, etc.

Table 2: Basic Statistical Features and Text Features.

3.4.2 LDA Topic Features

LDA topic model (Blei et al., 2003) is employed for **latent semantic analysis** of comment texts, mapping high-dimensional word space to low-dimensional topic space. Each comment is assigned to its highest-probability topic as a categorical feature (1D).

As shown in Figure 2 (see Appendix D) and Table 3, when text is duplicated, first-posted comments receive popularity **over an order of magnitude higher** than follower comments.

Post Order	Mean Reply	Mean Like
First Post	2.24	15.95
Follower	0.17	1.13
Ratio	13.5×	14.2×

Table 3: Popularity difference between first and follower posts.

Direct Jaccard similarity computation is $O(n^2)$. MinHash algorithm (Broder, 1997) achieves approximate computation through **Locality Sensitive Hashing (LSH)**, with retrieval examples provided in Figure 3 in the Appendix.

4 Evaluation Metrics

This paper establishes an “**accuracy-calibration dual-axis evaluation framework**”, comprehensively assessing model performance from two orthogonal dimensions: prediction accuracy and uncertainty calibration.

4.1 Prediction Accuracy Metrics

MSLE (Mean Squared Logarithmic Error):

$$\text{MSLE} = \frac{1}{n} \sum_{i=1}^n (\log(y_i + c) - \log(\hat{y}_i + c))^2 \quad (7)$$

ACP@(α, δ) (Accuracy within Tolerance):

$$\text{ACP} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[|y_i - \hat{y}_i| \leq \max(\alpha \cdot y_i, \delta)] \quad (8)$$

4.2 Uncertainty Calibration Metrics

Log-Scale NLL: See Equation 6.

PICP@95% (Prediction Interval Coverage Probability):

$$\text{PICP} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[y_i \in [\text{Lower}_i, \text{Upper}_i]] \quad (9)$$

MPIW (Mean Prediction Interval Width):

$$\text{MPIW} = \frac{1}{n} \sum_{i=1}^n (\text{Upper}_i - \text{Lower}_i) \quad (10)$$

5 Experiments

5.1 Dataset: Real-World Social Media Corpus

Our data is sourced from the Sina Weibo platform, with collection period from March 27 to April 14, 2025. Dataset statistics are shown in Table 4.

Dataset	Sample Count	Proportion
Training Set	217,234	80%
Validation Set	27,154	10%
Test Set	27,154	10%

Table 4: Dataset Statistics.

Model	MSLE	ACP	PICP	MPIW
NGBoost	0.0350	97.56%	94.30%	8.8M
BGE-Attn	0.0347	97.80%	97.75%	3.0050

Table 5: Comparison with NGBoost Baseline (Test Set).

5.2 Main Experimental Results

BGE-Attention coverage improved from 94.30% to 97.75% while MPIW reduced by **six orders of magnitude**.

5.3 Ablation Studies

Variant	MSLE	PICP	MPIW
Full Model	0.0347	97.75%	3.0050
w/o Dup	0.0364	97.21%	2.8602
w/o LogNLL	0.0391	98.14%	4.3090

Table 6: Ablation Study Results.

6 Conclusion

This paper proposes the BGE-Attention probabilistic prediction framework. Main conclusions are: (1) Reconstruction of prediction paradigm achieved Pareto optimality between accuracy and calibration; (2) Log-scale NLL loss function reconstructs scale invariance; (3) MinHash duplication features capture temporal position effects.

References

- Roja Bandari, Sitaram Asur, and Bernardo A Huberman. 2012. The pulse of news in social media: Forecasting popularity. In *Proceedings of ICWSM*, pages 26–33.
- David M Blei, Andrew Y Ng, and Michael I Jordan. 2003. Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022.
- Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. 2015. Weight uncertainty in neural networks. In *Proceedings of ICML*, pages 1613–1622.
- Andrei Z Broder. 1997. On the resemblance and containment of documents. In *Compression and Complexity of Sequences*, pages 21–29.

- Juan Deng and Xing Xie. 2020. Deep attention-based popularity prediction for social media. In *Proceedings of WWW*, pages 2822–2828.

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186.

- Tony Duan, Anand Avati, Daisy Yi Ding, and 1 others. 2020. NGBoost: Natural gradient boosting for probabilistic prediction. In *Proceedings of ICML*, pages 2690–2700.

- Yarin Gal. 2016. *Uncertainty in deep learning*. Ph.D. thesis, University of Cambridge.

- Yarin Gal and Zoubin Ghahramani. 2016. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of ICML*, pages 1050–1059.

- Bing Liu. 2019. *Sentiment analysis: Mining opinions, sentiments, and emotions*. Cambridge University Press.

- Yinhan Liu, Myle Ott, Naman Goyal, and 1 others. 2019. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.

- Chuhan Wu, Fangzhao Wu, Suyu Ge, and 1 others. 2019. Neural news recommendation with multi-head self-attention. In *Proceedings of EMNLP-IJCNLP*, pages 6389–6394.

- Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. 2023. C-pack: Packaged resources to advance general Chinese embedding. *arXiv preprint arXiv:2309.07597*.

- Jaewon Yang and Jure Leskovec. 2011. Patterns of temporal variation in online media. In *Proceedings of WSDM*, pages 177–186.

- Lan Zhang and Wei Zhang. 2020. Brand monitoring using social media analytics. *Journal of Marketing Research*, 57(4):741–762.

A High-Frequency User Statistics

Table 7 lists high-frequency users appearing ≥ 20 times in the dataset.

B Xiaomi-Related Vocabulary Lists

B.1 Feature Extraction Vocabulary

Xiaomi, SU7, Lei Jun, Intelligent driving, etc.

B.2 Special Embedding Keywords

Brand, Competitors, Smart Driving, etc.

No.	User	Count	No.	User	Count
1	Xiaomi Legal	328	11	I am Dabin	27
2	Lei Jun	292	12	Tech Xinyi	26
3	Xiaomi Auto	69	13	US Embassy	26
4	Wang Hua	66	14	AI Escape	24
5	AITO Legal	49	15	Xiaomi Inc	24
6	Schr. Cat	33	16	Alpaca Paj.	24
7	Yu Chengdong	32	17	Lu Weibing	22
8	Spokerspersion	29	18	Shiyu...	20
9	XiaoSuanMiao	29	19	Li Yunfei	20
10	Almighty Bear	27			

Table 7: High-Frequency Users and Their Occurrence Counts.

C Weibo Emoji List

The model uses 40 Weibo emojis: [doge], [haha], [laugh-cry], [allow-sad], [husky], etc.

D Additional Figures

G	H	I	J	K
用户昵称	评论内容	发布时间	子评论数	点赞数
身骑白马	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:06	1856	20471
傅然江木	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险公司，而是找汽车商家[	2025/4/1 23:22	0	3
超低牢雷	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:24	0	1
宝你铭	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:29	0	1
临江雪	是这样的/0身骑白马我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险	2025/4/1 23:57	0	0
伤口weuni	00000 我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商	2025/4/2 6:03	0	2
雷律66	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/2 6:42	54	429
肖康东	我的荣放有次跟车过近突然提醒我注意危险，我都没有发现有预碰撞提醒功能，丰田的销售t	2025/4/2 9:10	0	0
快_Ke	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的...	2025/4/2 11:45	1	4
感谢瞬间8	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/2 11:57	2	8

Figure 2: Comment text duplication patterns.

评论ID	评论内容	发布时间	子评论数	点赞数
10001	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:06	1856	20471
10002	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:22	0	3
10003	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:24	0	1
10004	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/1 23:29	0	1
10005	是这样的/0身骑白马我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险	2025/4/1 23:57	0	0
10006	00000 我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商	2025/4/2 6:03	0	2
10007	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/2 6:42	54	429
10008	我的荣放有次跟车过近突然提醒我注意危险，我都没有发现有预碰撞提醒功能，丰田的销售t	2025/4/2 9:10	0	0
10009	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的...	2025/4/2 11:45	1	4
10010	我这辈子第一次听说，出了交通事故在第一时间不找警察，不找保险，而是找汽车商家的	2025/4/2 11:57	2	8

Figure 3: MinHash retrieval example.